SAS Course - Homework 1

From Raw Data to Insights: EDA of the Diabetes Dataset in SAS Viya

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Dataset Overview

Dataset Structure

- **Observations:** 403 patients
- **Total Variables:** 19 initial features

Variables of Interest (Analyzed in EDA)

- **Demographics:**
  - Age (years)
  - Gender (male/female)
- **Biometrics:**
  - Height (inches) & Weight (pounds)
- **Clinical & Labs:**
  - Glycosylated Hemoglobin (%) – *Used for Diabetes classification*
  - Total Cholesterol (mg/dL)

Other Available Data

- Blood pressure (systolic/diastolic), HDL, body frame, location, etc.

Data Preprocessing

- **BMI Calculation**
  - Converted weight (pounds to kilograms) and height (inches to meters).
  - Applied formula: $BMI = \frac{weight}{height^2}$.
- **BMI Categorization**
  - Applied CDC guidelines (4 levels).
  - Grouped by: *Underweight*, *Normal Weight*, *Overweight*, *Obese*.
- **Diabetes Indicator**
  - Binary outcome from Glycosylated Hemoglobin (`glyhb`).
  - **Positive (1):** $\texttt{glyhb} > 6.5$ **Negative (0):** $0 \leq \texttt{glyhb} \leq 6.5$.
  - *Note:* Values ≤ 0 or structurally missing data remain missing.

Descriptive Distributions

Gender Distribution

- Female: 234 (**58.1%**)
- Male: 169 (41.9%)

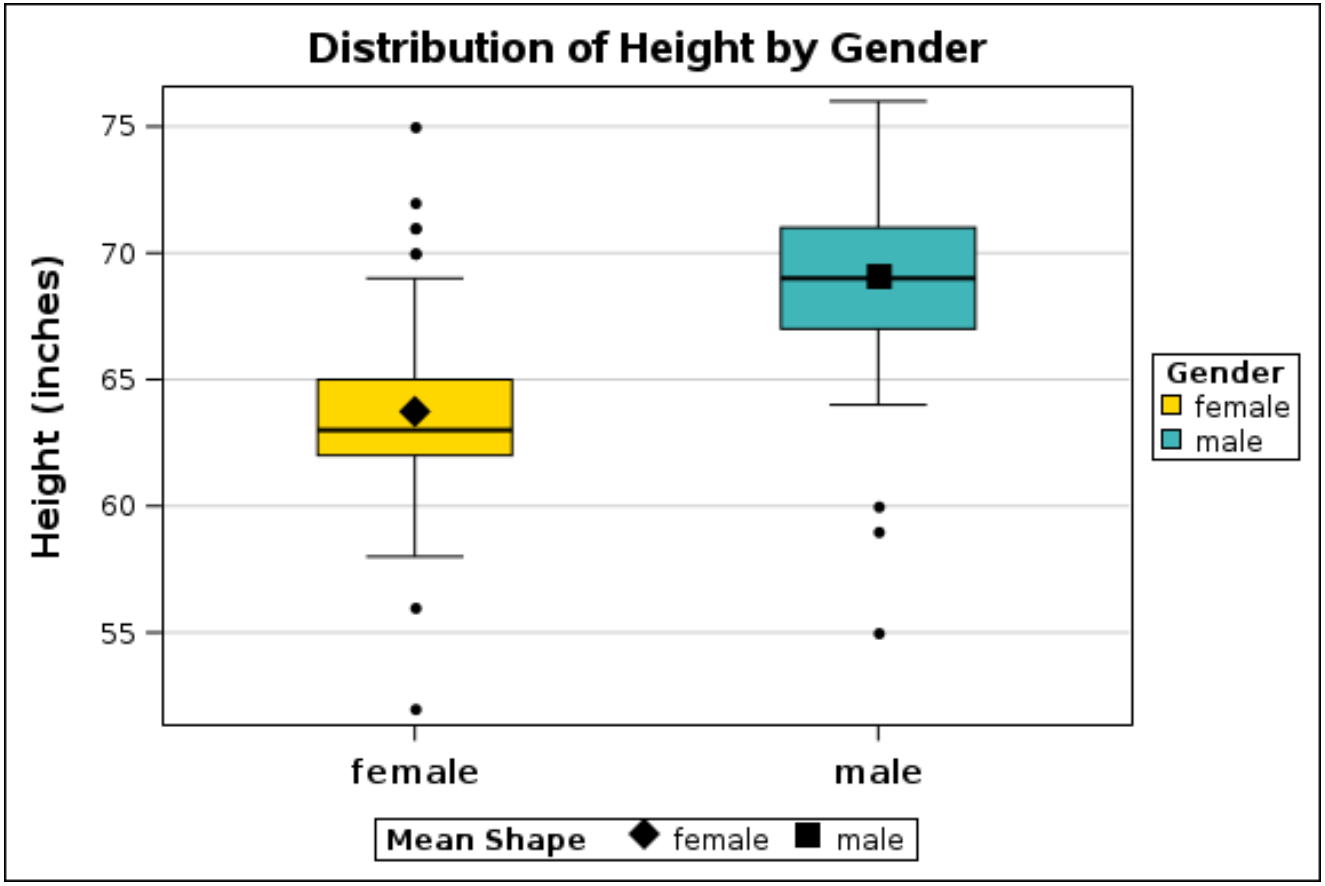
BMI Categories

- Underweight: 3.7%
- Normal: 28.0%
- Overweight: 30.5%
- Obese: **37.7%**

Key Insight

- Over **68.2%** of participants are classified as **Overweight or Obese**.
- The sample is notably **Female majority**.

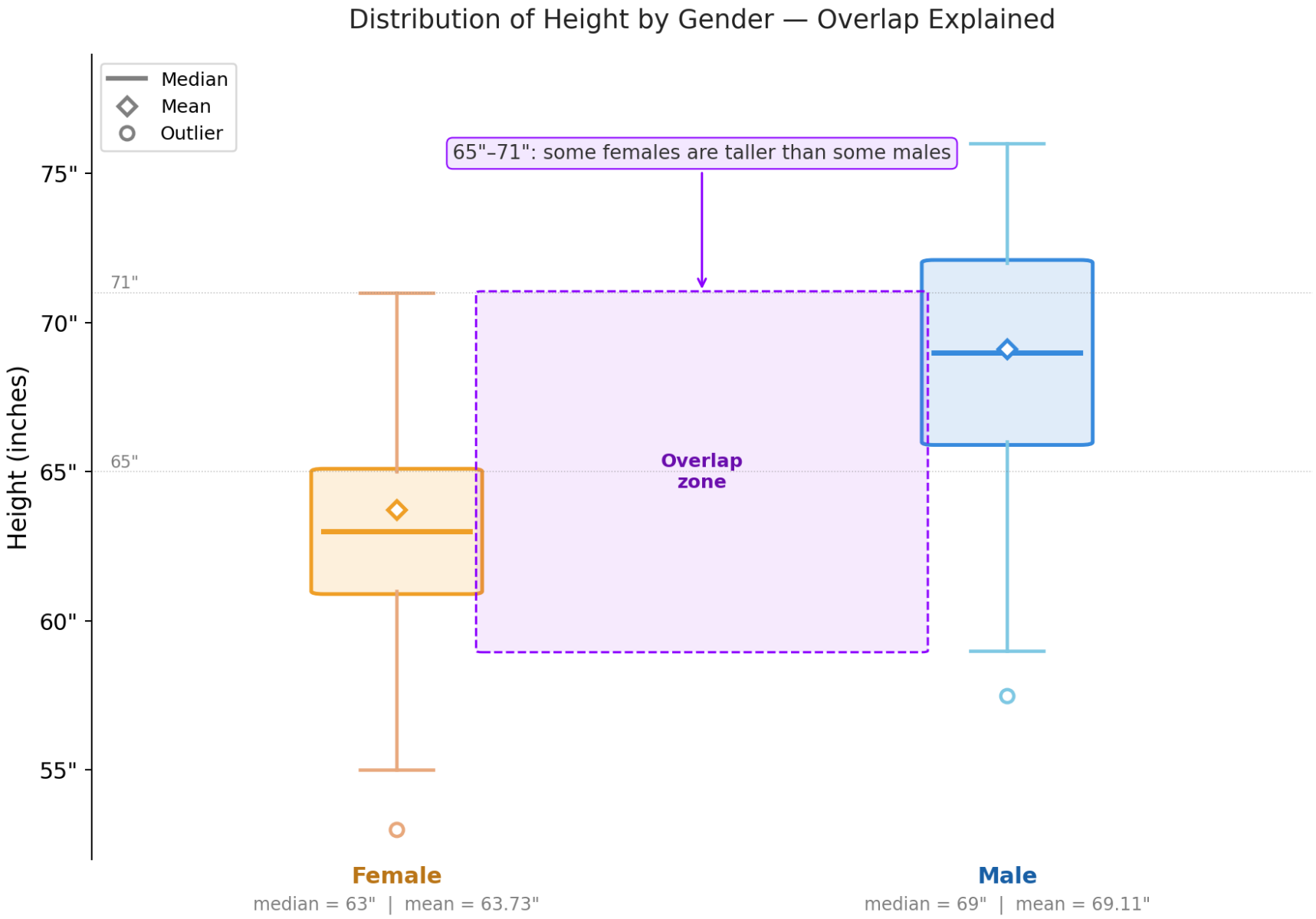
Distribution of Height by Gender



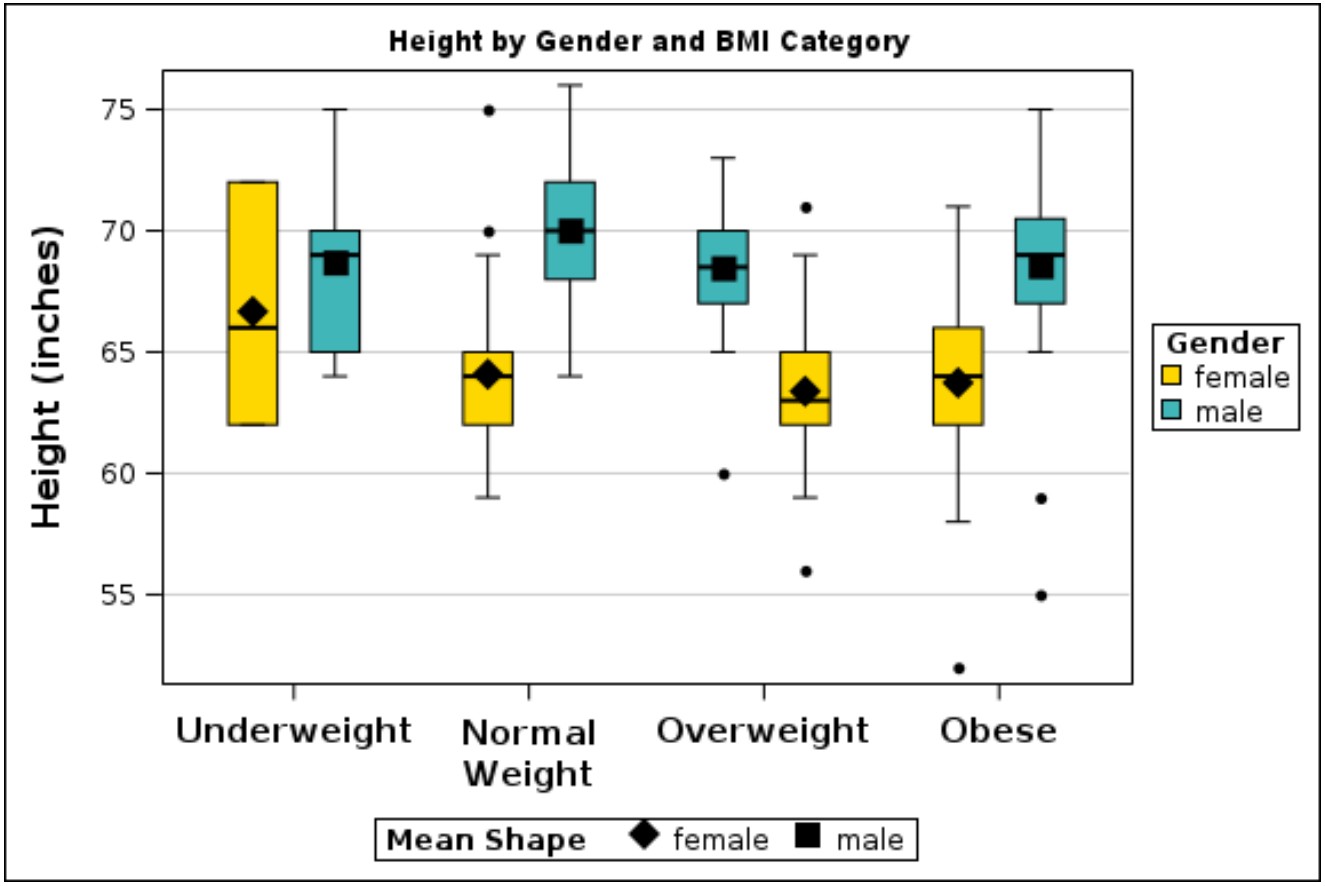
Key points:

- Higher median in males
- Slightly larger variability
- Overlapping distributions

But what is overlapping?



Height by BMI Category and Gender



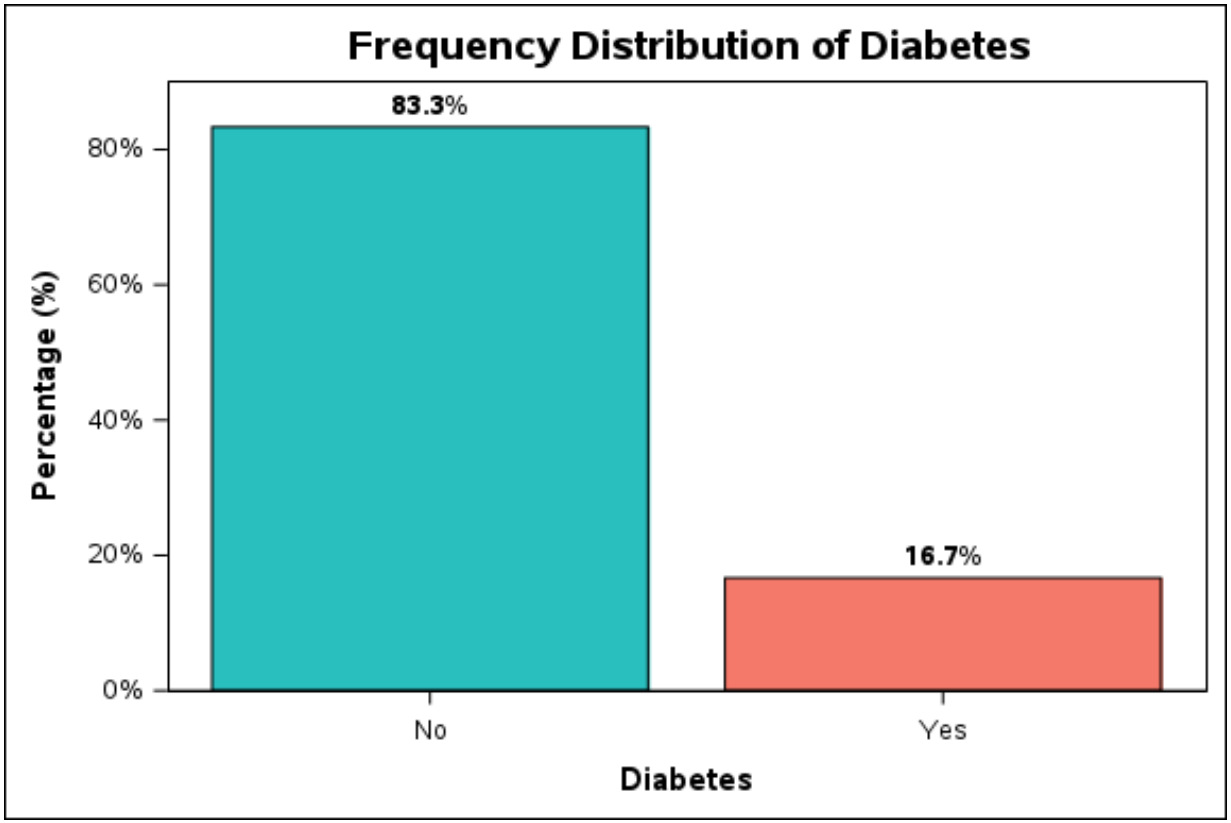
Key observations:

- Height differences persist across all BMI categories
- Males consistently taller than females
- Similar spread within BMI groups

Insight:

- BMI does not strongly affect height distribution

Distribution of Diabetes



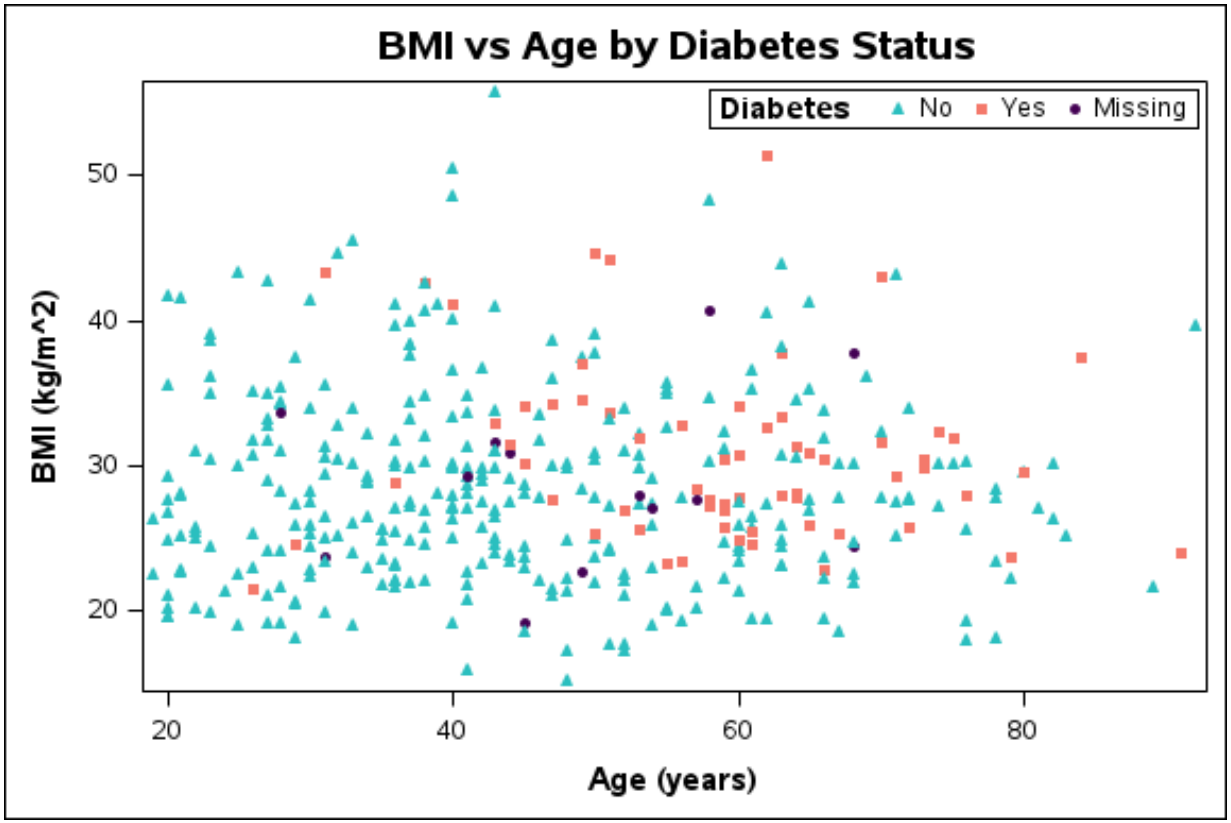
Key observations:

- Majority are non-diabetic (83.3%)
- Smaller proportion with diabetes (16.7%)

Insight:

- Moderate prevalence of diabetes in the sample

BMI vs Age by Diabetes Status



Observations:

- **No linear trend** is observed between age and BMI.
- **Non-diabetics** show wide variability across all ages and BMI.
- **Diabetics** concentrate at higher BMI (> 30) and older ages (> 45).

Clinical Insight:

- Higher BMI seems to be associated with diabetes, especially as age increases.

Comparison of Clinical Indicators (Means)

Table: Mean Values by Diabetes Status

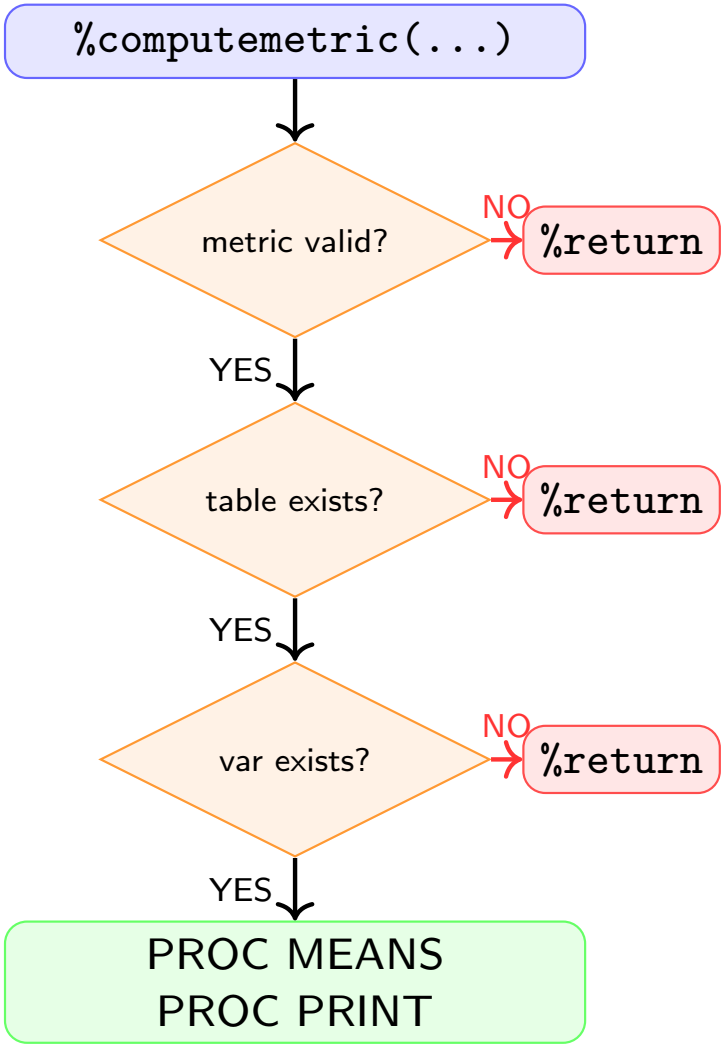
Variable	No Diabetes (0)	Diabetes (1)
Age (years)	44.4	58.4
BMI (kg/m^2)	28.4	30.8
Weight (lbs)	174.6	191.5
Cholesterol (mg/dL)	203.0	228.8
Glycosylated Hemoglobin (%)	4.7	9.9

Key Insights:

- **Age:** Diabetic patients are $\sim$14 years older on average.
- **Obesity:** The diabetes group mean BMI (30.8) falls into the Obese category.
- **Cholesterol:** Notably higher total cholesterol, indicating elevated cardiovascular risk.

Building the Custom SAS Macro

Validation Flow



Key Validation Logic

```
/* Validate metric */
%if %upcase(&metric) ne MEAN and
    %upcase(&metric) ne MEDIAN and
    %upcase(&metric) ne SUM %then %do;
  %put ERROR: Invalid metric.; %return;
%end;

/* Validate table exists */
%let dsid = %sysfunc(open(&table));
%if &dsid = 0 %then %do;
  %put ERROR: Table not found.; %return;
%end;

/* Validate variable exists */
%let varnum=%sysfunc(varnum(&dsid,&variable));
%if &varnum = 0 %then %do;
  %put ERROR: Var not found.; %return;
%end;
```

Why this matters

- **Early exit** — fails fast, no wasted computation
- **Modular** — stored separately in `mymacros.sas`
- **Reusable** across any table & variable

Executing the Macro & Final Results

1. Loading and Calling the Macro

Listing 1: Execution Example

```
/* Load the macro into the session */
%include "&classpath./hw1/mymacros.sas";

/* Example Call */
%computemetric(table=hw1.diab2, variable=height, metric=mean);
```

2. Computed Outputs

Dataset	Variable	Metric	Value
hw1.diab2	height	Mean	66.02
hw1.diab2	weight	Median	172.5
hw1.diab2	chol	Sum	183,554
sashelp.cars	MSRP	Mean	32,774.86

Thank You!

Any questions or comments?

Statistical Programming with SAS – Homework 1

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